

Jiahan Yang

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Education

Xi'an Jiaotong-Liverpool University

Sep 2024 – Jun 2028

B.Eng. in Mechatronics and Robotic Systems | WAM: 75+ / 100 (First Class, UK system)
Relevant coursework: Embedded Systems, Circuit Theory, Electronics, Control Systems

Research Interests

Robot learning · loco-manipulation · imitation learning · autonomous navigation · SLAM/GNSS fusion · world models for robotics

Selected Research & Robotics Projects

ATEC 2026 Simulation Challenge — Legged Loco-Manipulation

May 2026 – Present

Team “Zhu-Yang Farm” | Isaac Lab, RSL-RL / PPO, Unitree B2, AgileX Piper, Python

- Led the full technical development for a 2-member team, currently ranked **1st** by combined score across two tracks.
- Built an Isaac Lab control stack for a legged robot with an arm to autonomously navigate, detect, reach, pick, transport, and place randomly distributed objects in a 20 m × 20 m arena.
- Trained crouch-aware locomotion policies with a 4D command interface for linear velocity, yaw rate, and body-height control, enabling near-ground manipulation.
- Implemented a submission-ready pipeline combining RL locomotion, finite-state task planning, pure-Python Piper 6-DoF IK, HSV+depth detection, and corrected camera projection.

SO-101 Real-Robot Imitation Learning

Spring 2026

SO-101 Manipulator, $\pi 0.5$ Policies, Imitation Learning, Python

- Deployed $\pi 0.5$ imitation-learning policies on a physical SO-101 robotic arm for tabletop cleanup and small-cube grasping tasks.
- Collected and adapted demonstrations for real-world manipulation, focusing on action execution, calibration, grasp timing, and systematic failure analysis.

XJTLU Autonomous Vehicle Research Platform

2025 – Present

ROS 2 Humble, Jetson Orin NX, GNSS / RTK, Nav2 MPPI, FAST-LIO2, PGO, STM32

- Responsible for GNSS/RTK route navigation, Jetson deployment, and algorithm modification in a ROS 2 autonomous vehicle stack.
- Integrated route-navigation workflows around RTK GNSS, route collection, live alignment, waypoint execution, and Nav2 MPPI local planning.
- Worked with Livox MID-360 LiDAR, UM982 RTK, FAST-LIO2 odometry, PGO map correction, STM32 motor control, runtime logging, and launch automation on Jetson Orin NX.

ReKep-Based Supermarket Picking Simulation

2026

ReKep, Relational Keypoint Constraints, Language-Conditioned Manipulation

- Adapted ReKep-style relational keypoint constraints for a simulated supermarket item-picking scenario, studying language-conditioned manipulation through object-keypoint relations and spatial constraints.

Additional Engineering Experience

STM32, PCB Design, PID Control, Kalman Filtering, SolidWorks, ESP8266

- Built STM32 robotics prototypes including a smart desktop robot and a two-wheel self-balancing robot, covering PCB layout, soldering, servo / OLED control, PID attitude control, and Kalman-filter-based state estimation.

Technical Skills

Robotics	ROS 2, Isaac Lab, Nav2, FAST-LIO2, PGO, SLAM, GNSS/RTK, LiDAR/IMU integration, robot kinematics
Learning	PyTorch, reinforcement learning, imitation learning, VLA/robot policy reproduction, LangChain, LoRA fine-tuning
Programming	Python, C/C++, TypeScript, Linux, Git, Docker, FastAPI
Hardware	STM32, PCB design, serial/CAN communication, SolidWorks, Jetson deployment, sensor bring-up